CS 2601 Linear and Convex Optimization

9. Lagrange condition

Bo Jiang

John Hopcroft Center for Computer Science
Shanghai Jiao Tong University

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Outline

- Convex problems with equality constraints
- General equality constrained problems

Equality constrained convex problems

Consider the equality constrained convex optimization problem

$$\begin{aligned} \min_{\boldsymbol{x}} \quad & f(\boldsymbol{x}) \\ \text{s.t.} \quad & h_i(\boldsymbol{x}) = \boldsymbol{a}_i^T \boldsymbol{x} - b_i = 0, \quad i = 1, 2, \ldots, k \end{aligned}$$

where f is convex with $\mathrm{dom} f = \mathbb{R}^n$.

In a more compact form,

$$\begin{aligned} \min_{\boldsymbol{x}} \quad & f(\boldsymbol{x}) \\ \text{s.t.} \quad & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \end{aligned} \tag{CEC}$$

where $\boldsymbol{A}^T = [\boldsymbol{a}_1, \ldots, \boldsymbol{a}_k] \in \mathbb{R}^{n \times k}$, $\boldsymbol{b} = (b_1, \ldots, b_k) \in \mathbb{R}^k$.

We assume f is differentiable and the problem is feasible.

Feasible set and feasible directions

The feasible set is

$$X = \{\boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}\}$$

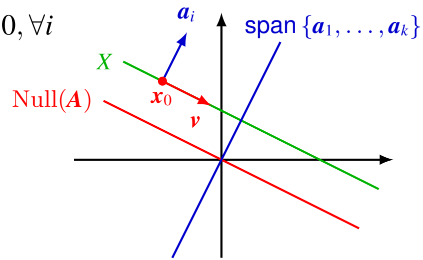
Given any $\boldsymbol{x}_0 \in X$,

$$X = \boldsymbol{x}_0 + \mathrm{Null}(\boldsymbol{A})$$

where $\mathrm{Null}(\boldsymbol{A}) = \{\boldsymbol{x} : \boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}\} = \{\boldsymbol{x} : \boldsymbol{a}_i^T\boldsymbol{x} = 0, i = 1, \ldots, k\}$ is the null space of $\boldsymbol{A}$.

$\mathrm{Null}(\boldsymbol{A})$ is precisely the set of feasible directions (at any $\boldsymbol{x}_0 \in X$)

$\boldsymbol{x}_0 + t\boldsymbol{v} \in X$ for small $t > 0 \iff \boldsymbol{a}_i^T\boldsymbol{v} = 0, \forall i$



Optimality condition

Lemma. If $\boldsymbol{x}^*$ is a minimum of (CEC), then

$$\nabla f(\boldsymbol{x}^*)^T \boldsymbol{v} \geq 0,$$

for all feasible direction $\boldsymbol{v}$, i.e. for $\boldsymbol{v} \in \mathrm{Null}(\boldsymbol{A})$.

Proof. Let $\boldsymbol{v} \in \mathrm{Null}(A)$ be a feasible direction. Then $\boldsymbol{x}^* + t\boldsymbol{v} \in X$ for all $t \geq 0$. Since $f(\boldsymbol{x}^* + t\boldsymbol{v}) \geq f(\boldsymbol{x}^*)$,the directional derivative

$$\lim_{t \downarrow 0} \frac{f(\boldsymbol{x}^* + t\boldsymbol{v}) - f(\boldsymbol{x}^*)}{t} = \nabla f(\boldsymbol{x}^*)^T \boldsymbol{v} \geq 0$$

Corollary. If $\boldsymbol{x}^*$ is a minimum of (CEC), then

$$\nabla f(\boldsymbol{x}^*)^T \boldsymbol{v} = 0,$$

for all $\boldsymbol{v} \in \mathrm{Null}(\boldsymbol{A})$, i.e. $\nabla f(\boldsymbol{x}^*) \perp \mathrm{Null}(\boldsymbol{A})$.

Proof. If $\boldsymbol{v} \in \mathrm{Null}(\boldsymbol{A})$, then $-\boldsymbol{v} \in \mathrm{Null}(\boldsymbol{A})$. Thus

$$\nabla f(\boldsymbol{x}^*)^T \boldsymbol{v} \geq 0, \nabla f(\boldsymbol{x}^*)^T (-\boldsymbol{v}) \geq 0 \implies \nabla f(\boldsymbol{x}^*)^T \boldsymbol{v} = 0.$$

Lagrange condition

Theorem. $\boldsymbol{x}^*$ is a global minimum of (CEC) iff

- $\boldsymbol{x}^*$ is feasible, i.e. $\boldsymbol{x}^* \in X$
- there exists $\boldsymbol{\lambda}^* = (\lambda_1^*, \ldots, \lambda_k^*) \in \mathbb{R}^k$ such that

$$\nabla f(\boldsymbol{x}^*) + \boldsymbol{A}^T \boldsymbol{\lambda}^* = \boldsymbol{0},$$

  or written out,

$$\nabla f(\boldsymbol{x}^*) + \sum_{i=1}^{k} \lambda_i^* \boldsymbol{a}_i = \boldsymbol{0}.$$

  The constants $\lambda_1^*, \ldots, \lambda_k^*$ are called Lagrange multipliers.

Recall $h_i(\boldsymbol{x}) = \boldsymbol{a}_i^T \boldsymbol{x} - b_i$, so $\boldsymbol{a}_i = \nabla h_i(\boldsymbol{x}^*)$. Then the above condition can be written as

$$\nabla f(\boldsymbol{x}^*) + \sum_{i=1}^{k} \lambda_i^* \nabla h_i(\boldsymbol{x}^*) = \boldsymbol{0}.$$

Lagrange condition (cont'd)

Proof. "$\Rightarrow$". Assume $\boldsymbol{x}^*$ is a global minimum.

- By the previous corollary,

$$\nabla f(\boldsymbol{x}^*) \perp \mathrm{Null}(\boldsymbol{A})$$

  so

$$\nabla f(\boldsymbol{x}^*) \in \mathrm{Null}(\boldsymbol{A})^\perp$$

- Since $\mathrm{Null}(\boldsymbol{A})^\perp = \mathrm{Range}(\boldsymbol{A}^T)$, so there exists $-\boldsymbol{\lambda}^*$ s.t.

$$\nabla f(\boldsymbol{x}^*) = -\boldsymbol{A}^T\boldsymbol{\lambda}^*$$

"$\Leftarrow$". Assume $\boldsymbol{x}^* \in X$ and $\nabla f(\boldsymbol{x}^*) + \boldsymbol{A}^T\boldsymbol{\lambda}^* = \mathbf{0}$ for some $\boldsymbol{\lambda}^*$.

- By the first-order condition for convexity, for any $\boldsymbol{x} \in X$,

$$\begin{aligned} f(\boldsymbol{x}) &\geq f(\boldsymbol{x}^*) + \nabla f(\boldsymbol{x}^*)^T(\boldsymbol{x} - \boldsymbol{x}^*) \\ &= f(\boldsymbol{x}^*) + (\boldsymbol{\lambda}^*)^T\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{x}^*) \\ &= f(\boldsymbol{x}^*), \qquad \boldsymbol{A}\boldsymbol{x} = \boldsymbol{A}\boldsymbol{x}^* = \boldsymbol{b} \text{ by feasibility} \end{aligned}$$

Lagrange condition (cont'd)

Define Lagrangian (or Lagrange function) by

$$\mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}) = f(\boldsymbol{x}) + \boldsymbol{\lambda}^T(\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}) = f(\boldsymbol{x}) + \sum_{i=1}^{k} \lambda_i(\boldsymbol{a}_i^T \boldsymbol{x} - b_i)$$

The optimality condition becomes the following Lagrange condition, aka KKT equations[1]

$$\begin{cases} \nabla_{\boldsymbol{x}} \mathcal{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*) = \nabla f(\boldsymbol{x}^*) + \boldsymbol{A}^T \boldsymbol{\lambda}^* = \boldsymbol{0} \\ \nabla_{\boldsymbol{\lambda}} \mathcal{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*) = \boldsymbol{A}\boldsymbol{x}^* - \boldsymbol{b} = \boldsymbol{0} \end{cases}$$

where $\nabla_{\boldsymbol{x}}$ and $\nabla_{\boldsymbol{\lambda}}$ are partial gradient w.r.t. $\boldsymbol{x}$ and $\boldsymbol{\lambda}$, or

$$\nabla \mathcal{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*) = \boldsymbol{0}$$

i.e. $(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)$ is a stationary point of $\mathcal{L}$.

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[1] KKT stands for Karush-Kuhn-Tucker. We'll see later why it is called as such.

Example

Consider

$$\min_{x_1,x_2} \quad f(x_1,x_2) = \frac{1}{2}x_1^2 + \frac{1}{2}x_2^2$$
$$\text{s.t.} \quad x_1 + 2x_2 = 1$$

Method 1. Reduction to an equivalent unconstrained problem.

$$g(x_2) \triangleq f(1-2x_2, x_2) = \frac{1}{2}(1-2x_2)^2 + \frac{1}{2}x_2^2$$

$$\min_{x_2} g(x_2) \implies g'(x_2^*) = 0 \implies x_2^* = \frac{2}{5} \implies x_1^* = 1 - 2x_2^* = \frac{1}{5}$$

Method 2. Lagrangian multipliers method. The Lagrangian is

$$\mathcal{L}(x_1, x_2, \lambda) = \frac{1}{2}x_1^2 + \frac{1}{2}x_2^2 + \lambda(x_1 + 2x_2 - 1)$$

By the Lagrange condition,

$$\begin{cases} \frac{\partial \mathcal{L}}{\partial x_1} = x_1 + \lambda = 0 \\ \frac{\partial \mathcal{L}}{\partial x_2} = x_2 + 2\lambda = 0 \\ \frac{\partial \mathcal{L}}{\partial \lambda} = x_1 + 2x_2 - 1 = 0 \end{cases} \implies \begin{cases} x_1^* = \frac{1}{5} \\ x_2^* = \frac{2}{5} \\ \lambda^* = -\frac{1}{5} \end{cases}$$

Example (cont'd)

$$\begin{aligned} \min_{x_1, x_2} \quad & f(x_1, x_2) = \frac{1}{2}x_1^2 + \frac{1}{2}x_2^2 \\ \text{s.t.} \quad & x_1 + 2x_2 = 1 \end{aligned}$$

- normal vector to the feasible set X

$$\boldsymbol{a} = (1, 2)$$

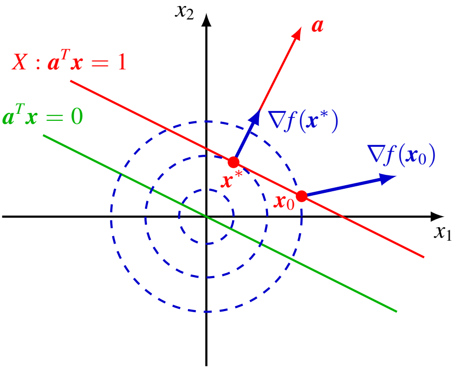
- gradient

$$\nabla f(\boldsymbol{x}) = \boldsymbol{x}$$

- at $\boldsymbol{x}^*$,

$$\nabla f(\boldsymbol{x}^*) = -\lambda^* \boldsymbol{a} \perp X$$

Note X is parallel to $\mathrm{Null}(\boldsymbol{a}^T)$.



Example

$$\begin{aligned} \min_{\boldsymbol{x}} \quad & f(\boldsymbol{x}) = \frac{1}{2}\|\boldsymbol{x}\|^2, \quad \text{where } \boldsymbol{A} = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 2 & 1 \end{bmatrix}, \boldsymbol{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ \text{s.t.} \quad & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \end{aligned}$$

Method 1. Reduction to an equivalent unconstrained problem.

- $\operatorname{rank} \boldsymbol{A} = 2$. Find two independent columns of $\boldsymbol{A}$, e.g. the first and third columns, and solve for the corresponding x_i's in terms of the others. Let $\boldsymbol{A}_1 = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, \boldsymbol{A}_2 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$. The constraints become

$$\boldsymbol{A}_1 \begin{bmatrix} x_1 \\ x_3 \end{bmatrix} + \boldsymbol{A}_2 x_2 = \boldsymbol{b} \implies \begin{bmatrix} x_1 \\ x_3 \end{bmatrix} = \boldsymbol{A}_1^{-1}\boldsymbol{b} - \boldsymbol{A}_1^{-1}\boldsymbol{A}_2 x_2 = \begin{bmatrix} 1 - 2x_2 \\ 2x_2 - 1 \end{bmatrix}$$

- Substitution into f yields

$$g(x_2) = f(1 - 2x_2, x_2, 2x_2 - 1) = (2x_2 - 1)^2 + \frac{1}{2}x_2^2 \implies x_2^* = \frac{4}{9}$$

- $x_1^* = 1 - 2x_2^* = \frac{1}{9}$, $x_3^* = 2x_2^* - 1 = -\frac{1}{9}$

Example (cont'd)

$$\min_{\boldsymbol{x}} \quad f(\boldsymbol{x}) = \frac{1}{2}\|\boldsymbol{x}\|^2, \qquad \text{where } \boldsymbol{A} = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 2 & 1 \end{bmatrix}, \boldsymbol{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$
$$\text{s.t.} \quad \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$$

Method 2. Lagrange multipliers method.

- The Lagrangian is

$$\mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}) = \frac{1}{2}\|\boldsymbol{x}\|^2 + \boldsymbol{\lambda}^T(\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b})$$

- Lagrange condition

$$\begin{cases} \nabla_{\boldsymbol{x}}\mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}) = \boldsymbol{x} + \boldsymbol{A}^T\boldsymbol{\lambda} = \boldsymbol{0} \\ \nabla_{\boldsymbol{\lambda}}\mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}) = \boldsymbol{A}\boldsymbol{x} - \boldsymbol{b} = \boldsymbol{0} \end{cases} \quad \text{or} \quad \begin{bmatrix} \boldsymbol{I} & \boldsymbol{A}^T \\ \boldsymbol{A} & \boldsymbol{O} \end{bmatrix} \begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{\lambda} \end{bmatrix} = \begin{bmatrix} \boldsymbol{0} \\ \boldsymbol{b} \end{bmatrix}$$

- Solve for $\boldsymbol{x}, \boldsymbol{\lambda}$ e.g. by substitution or block Gaussian elimination,

$$\begin{cases} \boldsymbol{x}^* = -\boldsymbol{A}^T\boldsymbol{\lambda}^* = \boldsymbol{A}^T(\boldsymbol{A}\boldsymbol{A}^T)^{-1}\boldsymbol{b} \\ \boldsymbol{\lambda}^* = -(\boldsymbol{A}\boldsymbol{A}^T)^{-1}\boldsymbol{b} \end{cases} \implies \begin{cases} \boldsymbol{x}^* = (\frac{1}{9}, \frac{4}{9}, -\frac{1}{9}) \\ \boldsymbol{\lambda}^* = (-\frac{1}{3}, \frac{1}{9}) \end{cases}$$

Example (cont'd)

Block Gaussian elimination.

- The augmented matrix is

$$\begin{bmatrix} \boldsymbol{I} & \boldsymbol{A}^T & \boldsymbol{0} \\ \boldsymbol{A} & \boldsymbol{O} & \boldsymbol{b} \end{bmatrix}$$

- Left multiply the first "row" by $-\boldsymbol{A}$ and add to the second "row",

$$\begin{bmatrix} \boldsymbol{I} & \boldsymbol{A}^T & \boldsymbol{0} \\ \boldsymbol{O} & -\boldsymbol{A}\boldsymbol{A}^T & \boldsymbol{b} \end{bmatrix}$$

- Left multiply the second "row" by $-(\boldsymbol{A}\boldsymbol{A}^T)^{-1}$ (why invertible?),

$$\begin{bmatrix} \boldsymbol{I} & \boldsymbol{A}^T & \boldsymbol{0} \\ \boldsymbol{O} & \boldsymbol{I} & -(\boldsymbol{A}\boldsymbol{A}^T)^{-1}\boldsymbol{b} \end{bmatrix}$$

- Left multiply the second "row" by $-\boldsymbol{A}^T$ and add to the first "row",

$$\begin{bmatrix} \boldsymbol{I} & \boldsymbol{O} & \boldsymbol{A}^T(\boldsymbol{A}\boldsymbol{A}^T)^{-1}\boldsymbol{b} \\ \boldsymbol{O} & \boldsymbol{I} & -(\boldsymbol{A}\boldsymbol{A}^T)^{-1}\boldsymbol{b} \end{bmatrix}$$

Example (cont'd)

$$\min_{\boldsymbol{x}} \quad f(\boldsymbol{x}) = \frac{1}{2}\|\boldsymbol{x}\|^2, \qquad \text{where } \boldsymbol{A} = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 2 & 1 \end{bmatrix}, \boldsymbol{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$
$$\text{s.t.} \quad \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$$

- vectors normal to the feasible set X

$$\text{span}\,\{\boldsymbol{a}_1, \boldsymbol{a}_2\}$$

  with $\boldsymbol{a}_1 = (1, 2, 0)$, $\boldsymbol{a}_2 = (2, 2, 1)$.

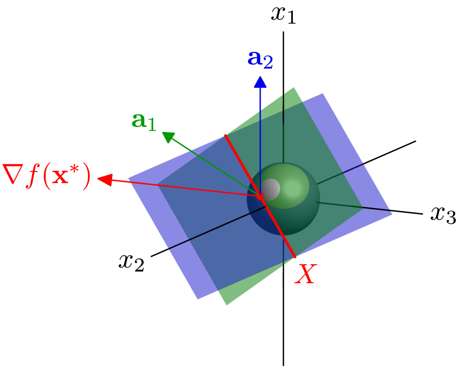
- gradient

$$\nabla f(\boldsymbol{x}) = \boldsymbol{x}$$

- at $\boldsymbol{x}^*$,

$$\nabla f(\boldsymbol{x}^*) = -\lambda_1^* \boldsymbol{a}_1 - \lambda_2^* \boldsymbol{a}_2 \perp X$$

  Note X is parallel to $\mathrm{Null}(\boldsymbol{A}^T)$.



Outline

- Convex problems with equality constraints
- General equality constrained problems